

Epistemic Dynamics: Deriving the Schrödinger Equation from the Principle of Least Surprise

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The interpretation of the quantum wavefunction Ψ remains a subject of debate. The "QBism" (Quantum Bayesianism) framework posits that Ψ represents an agent's subjective degrees of belief rather than an element of physical reality. In this paper, we formalize this view by deriving the Schrödinger equation from an information-theoretic variational principle. We demonstrate that if a probability density evolves to minimize a functional composed of Fisher Information (Parsimony) and Expected Energy (Accuracy), the dynamics are necessarily governed by the Schrödinger equation. This suggests that quantum mechanics is the unique statistical theory of a "self-evidencing" system minimizing prediction error under uncertainty, aligning with the "Information" component of the PCIR framework.

I. INTRODUCTION

Why is the universe quantum mechanical? Standard approaches treat the Schrödinger equation as a fundamental axiom. However, the "It from Qubit" program [1] and approaches based on Fisher Information [2, 3] suggest that physical laws may emerge from the rules of data processing.

The PCIR framework posits that physical systems minimize a free energy functional. In the context of dynamics, this implies a trade-off between maximizing epistemic accuracy and minimizing model complexity.

In this paper, we show that the kinetic energy term in the Schrödinger equation is isomorphic to the Fisher Information of the probability density. Consequently, the time-evolution of the wavefunction is a diffusion process on a statistical manifold, driven by the requirement to maintain optimal inference.

II. FISHER INFORMATION AS EPISTEMIC COST

Consider a particle described by a probability density function (PDF) $\rho(\mathbf{x}, t)$. The position $\mathbf{x}$ is a random variable. The "sharpness" or quality of this distribution is measured by the Fisher Information $\mathcal{I}$:

$$\mathcal{I}[\rho] = \int \frac{1}{\rho} (\nabla \rho)^2 d^3x = \int \rho (\nabla \ln \rho)^2 d^3x \quad (1)$$

Fisher Information measures the sensitivity of the distribution to translation. High $\mathcal{I}$ implies a highly localized (sharp) distribution. In the PCIR framework, high localization represents high **Complexity** (Parsimony cost) because it requires more bits to specify the coordinate.

III. THE VARIATIONAL DERIVATION

We postulate an action principle based on the trade-off between Knowledge (Fisher Information) and Constraint (Potential Energy).

A. The Madelung Transformation

We introduce a complex amplitude $\psi(\mathbf{x}, t)$ such that $\rho = |\psi|^2$. Following the Madelung decomposition:

$$\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{iS(\mathbf{x}, t)/\hbar} \quad (2)$$

where S is a phase factor (action).

B. The Information Action

We construct a Lagrangian $\mathcal{L}$ that minimizes the Fisher Information (Parsimony) subject to the constraint of the external potential $V(\mathbf{x})$ (Context/Environment). The simplest non-trivial information Lagrangian is:

$$\mathcal{L} = \underbrace{\frac{\hbar^2}{8m} \mathcal{I}[\rho]}_{\text{Epistemic Cost}} + \underbrace{\langle V \rangle}_{\text{External Cost}} - \lambda \underbrace{\left(\int \rho d^3x - 1 \right)}_{\text{Normalization}} \quad (3)$$

Remarkably, substituting the definition of $\mathcal{I}$ and ψ , this Lagrangian is equivalent to the expectation value of the Hamiltonian:

$$\int \mathcal{L} dt = \int dt \int d^3x \psi^* \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi \quad (4)$$

This is the standard Lagrangian for the Schrödinger field.

C. Equations of Motion

Applying the Euler-Lagrange equations $\frac{\delta \mathcal{L}}{\delta \psi^*} = 0$ yields:

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi \quad (5)$$

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Thus, the Schrödinger equation is the necessary equation of motion for a probability field that minimizes the sum of its Fisher Information (kinetic energy) and Potential Energy.

IV. MEASUREMENT AS BAYESIAN UPDATING

If Ψ is a probability amplitude derived from information constraints, then "collapse" is simply Bayesian updating.

Let H be the hypothesis (state of the system) and D be the observed data (measurement result). Bayes' theorem states:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)} \quad (6)$$

In Quantum Mechanics, the "Born Rule" $P = |\psi|^2$ provides the likelihood. The discontinuity of wavefunction collapse is identical to the discontinuity of a probability distribution when new information is acquired. It is an update of the *agent's knowledge* (the "Information" component of PCIR), not a physical contraction of a material wave.

[Image of Bayesian updating process]

V. DISCUSSION

We have established the **"I" (Information)** component of the PCIR framework.

1. **Kinetic Energy is Fisher Information:** The energy cost of a quantum state is the cost of its localizability (Parsimony).
2. **Dynamics are Inference:** The Schrödinger equation describes the optimal evolution of beliefs in a fluctuating environment.
3. **Collapse is Updating:** Measurement is the resolution of ambiguity (minimizing entropy).

This complements Paper 1 (Gravity as Fisher Metric) and Paper 2 (Time as RG Flow). Together, they suggest a universe where Geometry, Time, and Quantum States are all emergent features of a single information-processing principle.

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